Build Your Next Apache Spark Job in .NET Using Mobius

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SPARK SUMMIT EUROPE 2016

Mobius

C# API for building Apache Spark applications in .NET

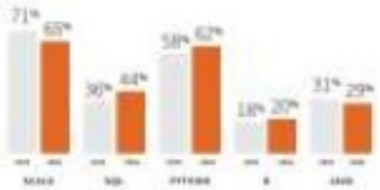
Motivation

- Enable organizations invested deeply in .NET to build Apache Spark applications in C#
- Reuse of existing .NET libraries in Spark applications

Yet Another Language Binding?

Spark Survey Results

Spark Users Employ Multiple Languages



MOST IMPORTANT ASPECTS OF SPARK

77%

Ease of programming

2015

76%

EASE OF PROGRAMMING

2016

FASTEST GROWING AREAS FROM 2014 TO 2015

+283% increase in Windows users

(went from 6% to 23% of users)

2015

WINDOWS USERS IN DEVELOPMENT

+39%

2015	2016
23%	32%

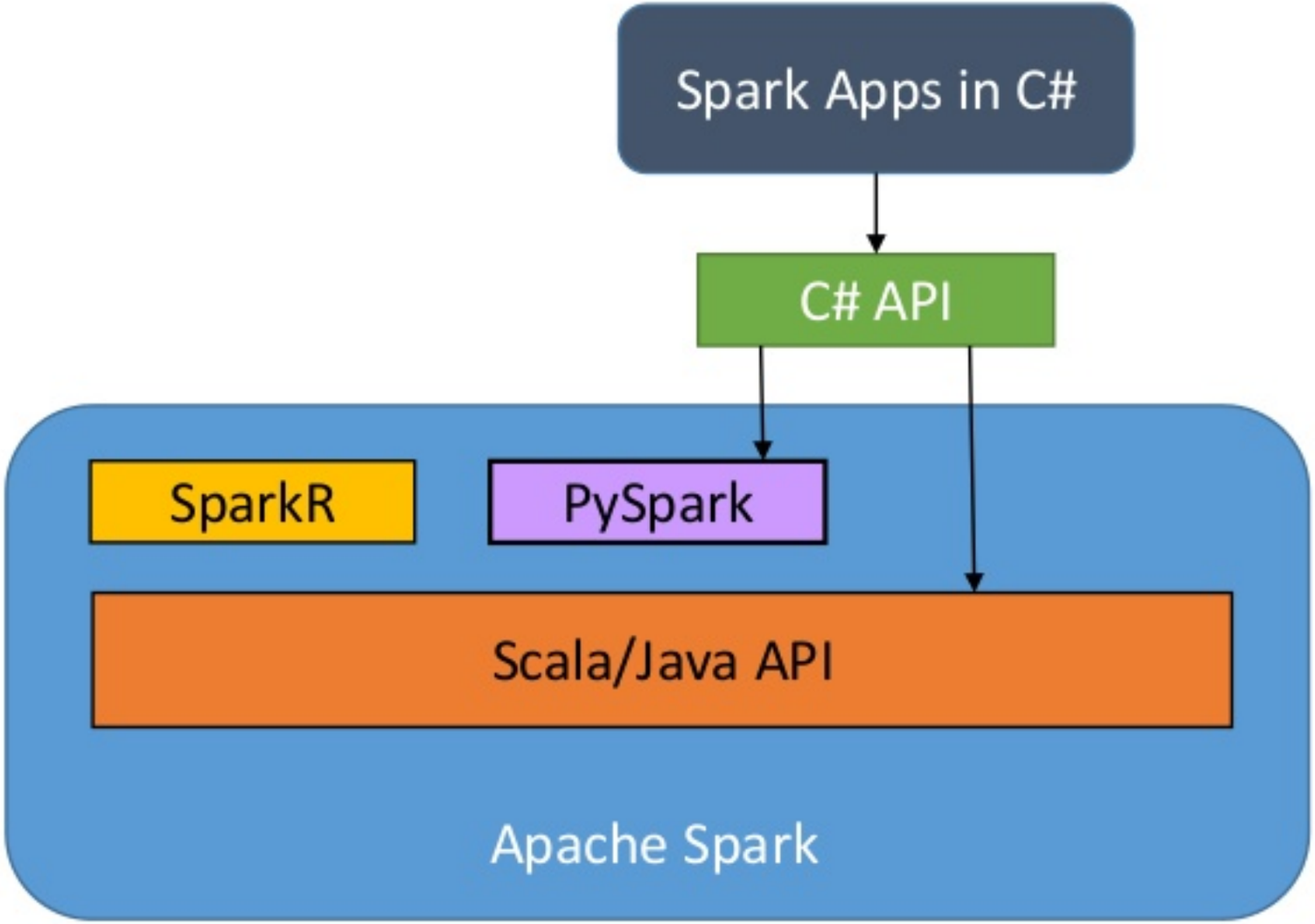
2016

Popularity of C#

- StackOverflow.com Developer Survey
- RedMonk Programming Language Rankings

.NET ecosystem ~ enabling languages like F#

Mobius & Spark



Word Count

Scala

```scala
val textFile = spark.textFile("hdfs://...")
val counts = textFile.flatMap(line => line.split(" "))
                     .map(word => (word, 1))
                     .reduceByKey(_ + _)
counts.saveAsTextFile("hdfs://...")
```

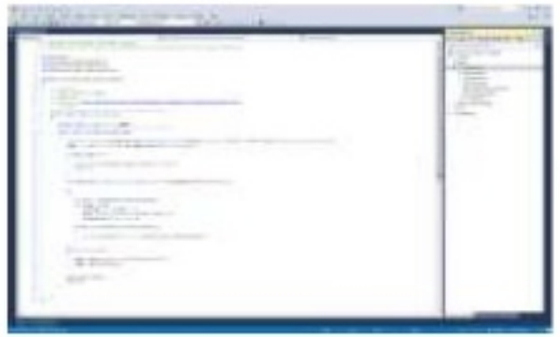
C#

```csharp
var textFile = sparkContext.TextFile(@"hdfs://...");
var counts = textFile.FlatMap(x => x.Split(' '))
                     .Map(w => new KeyValuePair<string, int>(w, 1))
                     .ReduceByKey((x, y) => x + y)
                     .Map(wordCount => $"{wordCount.Key},{wordCount.Value}");
counts.SaveAsTextFile(@"hdfs://...");
```

F#

```fsharp
let textFile = sparkContext.TextFile "hdfs://..."
let counts =
    textFile.FlatMap(fun x -> x.Split ' ' :> _)
          .Map(fun w -> KeyValuePair(w, 1))
          .ReduceByKey(fun x y -> x + y)
          .Map(fun wordCount -> sprintf "%s,%d" wordCount.Key wordCount.Value)
counts.SaveAsTextFile "hdfs://..."
```

Develop & Launch Mobius Applications



1 Add Reference to Mobius package in NuGet

2 Develop, debug, test Mobius driver application

3 Build Mobius driver

Get Mobius release

A

Get Mobius driver and dependencies

B

Run sparkclr-submit.cmd or sparkclr-submit.sh

C

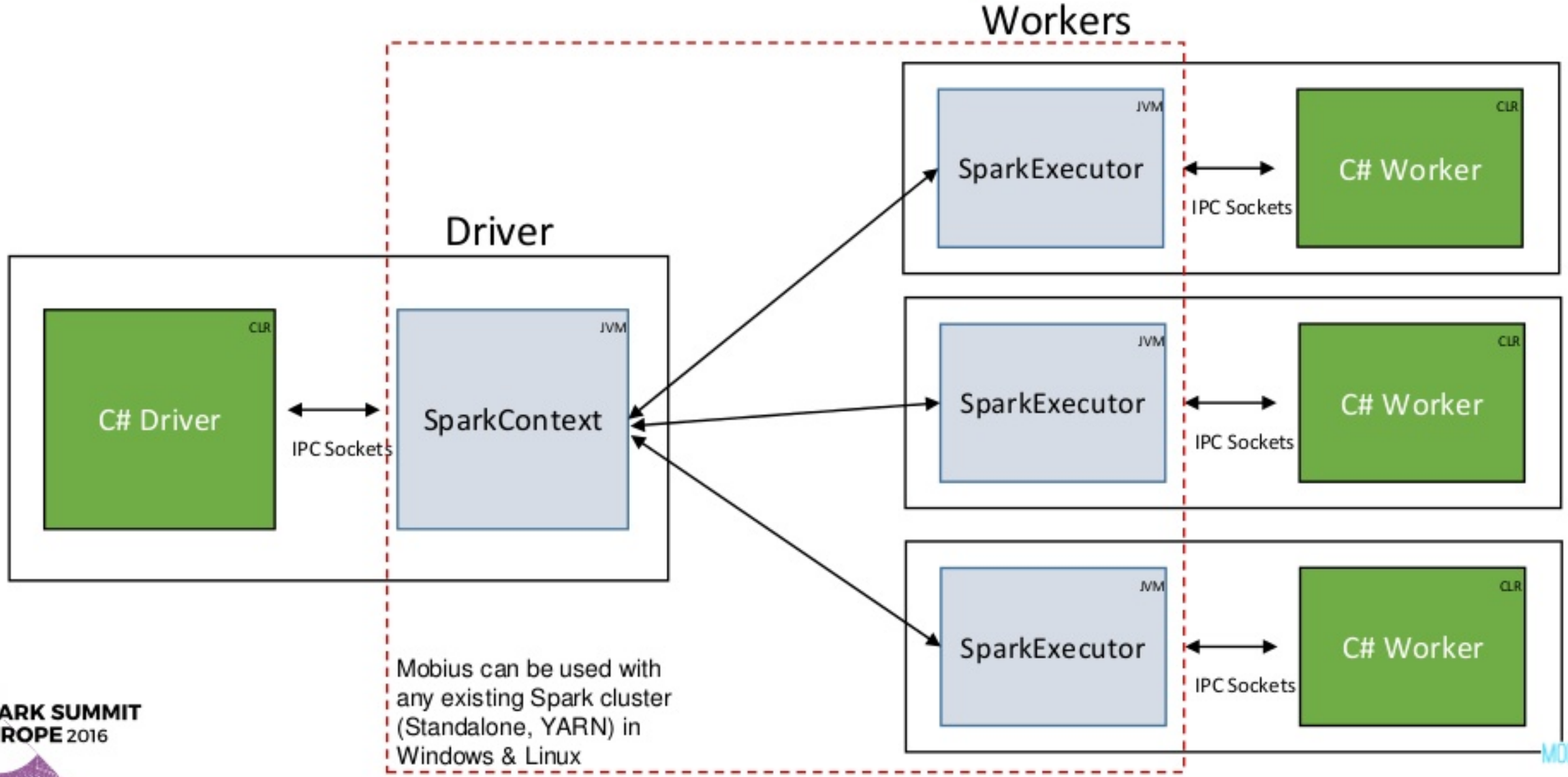
Runs Spark job

Spark Client

Example: sparkclr-submit.cmd

```
--master spark://IP:PORT
--total-executor-cores 200
--executor-memory 12g
--exe Pi.exe D:\Mobius\examples\Pi
```

Mobius & Spark

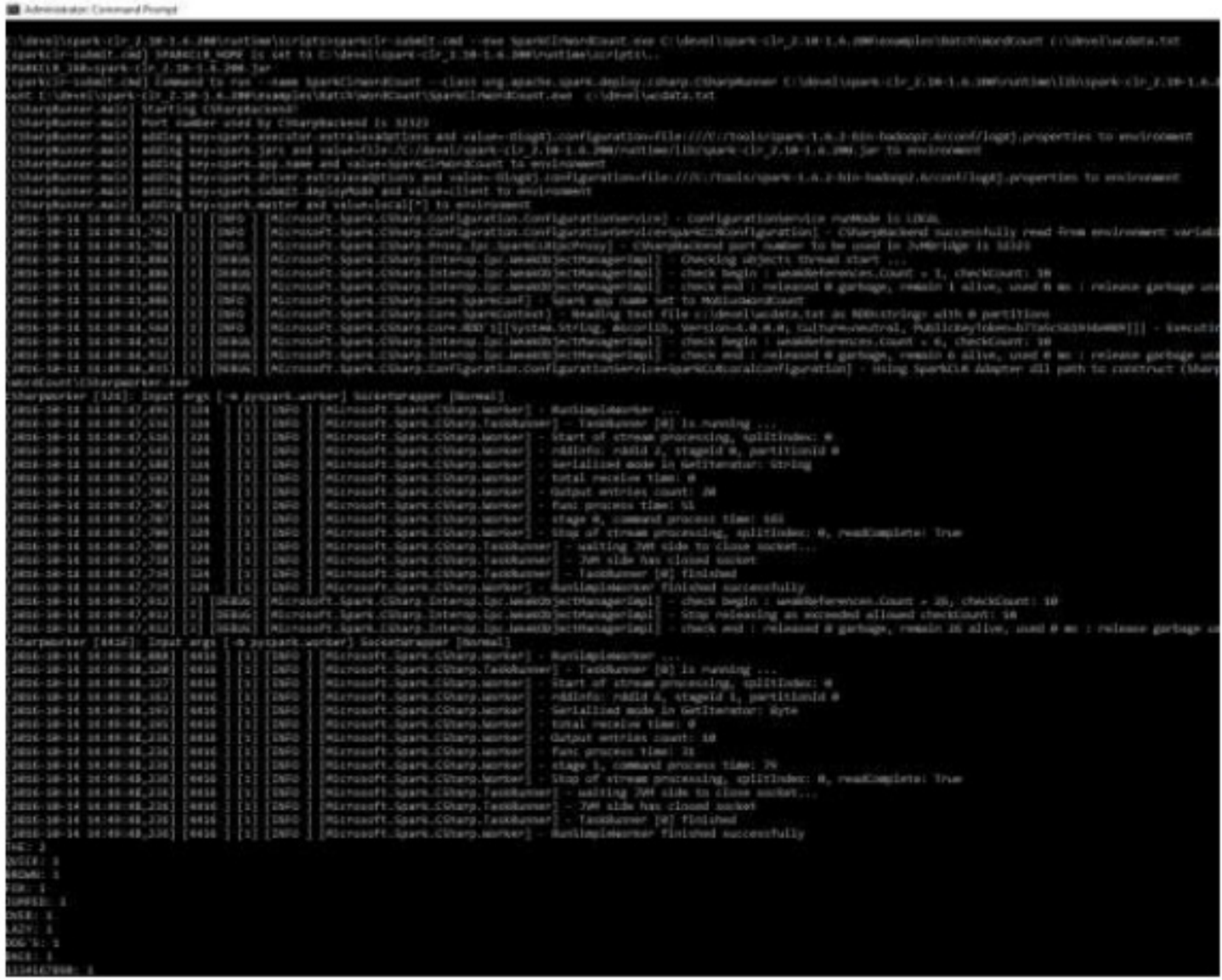


DEMO – RUNNING MOBIUS APP

sparkclr-submit.cmd

```
> sparkclr-submit.cmd
        --exe SparkClrWordCount.exe
        C:\spark-clr_2.10-1.6.200\examples\Batch\WordCount
        C:\temp\wcdata.txt
```

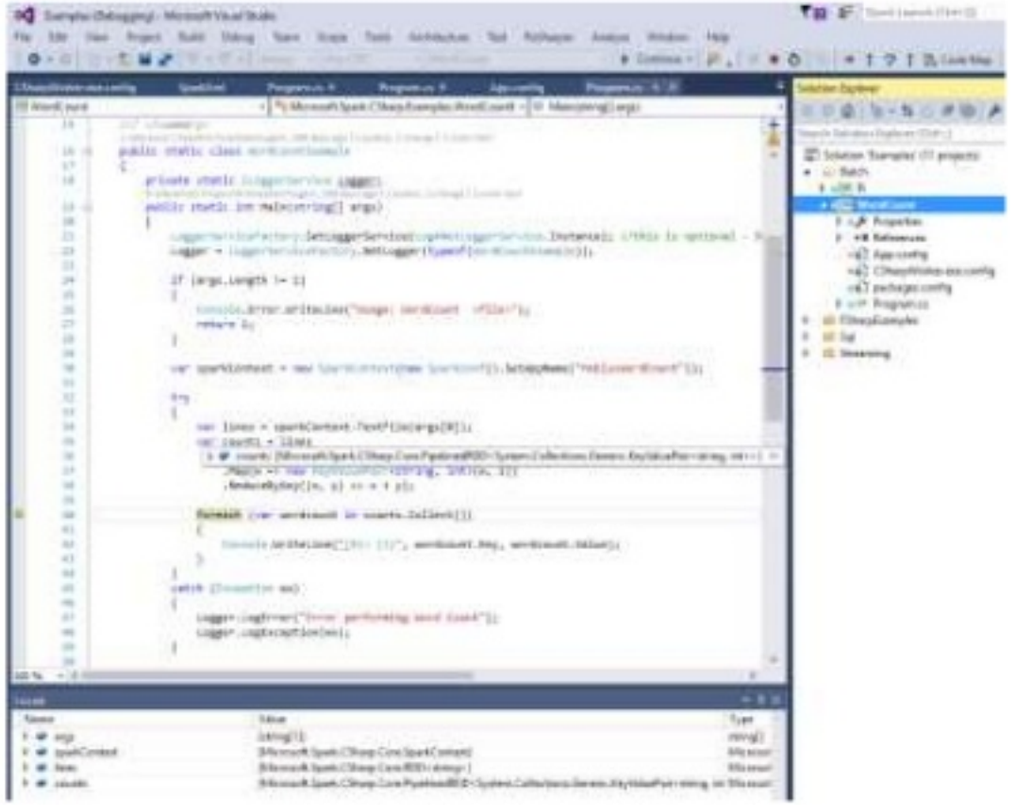
sparkclr-submit.sh in Linux



DEMO – DEBUGGING MOBIUS APP

Debug Mode

```
> sparkclr-submit.cmd debug
```



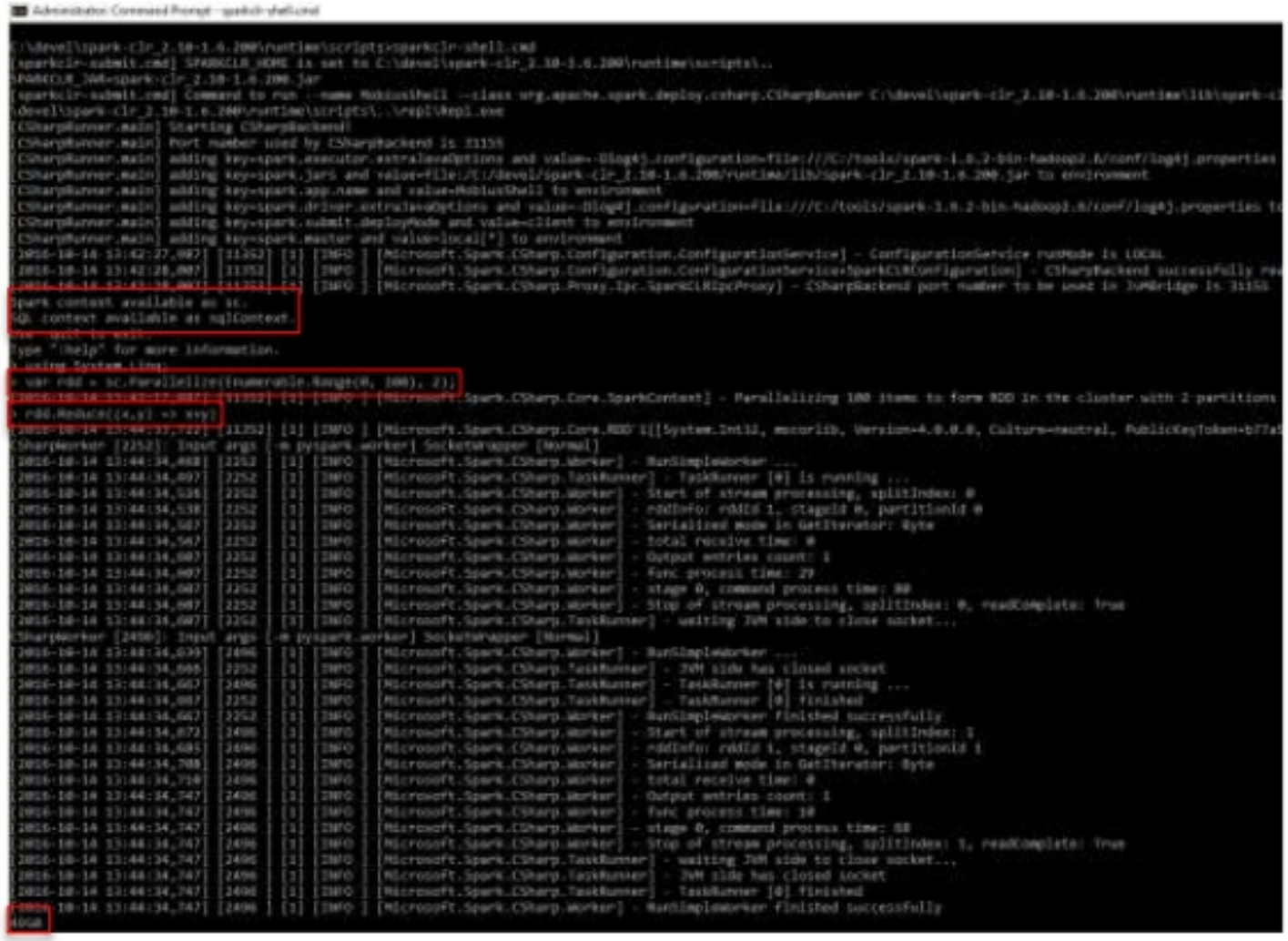
Debug Mobius Word Count Example in VS

(set CSharpWorkerPath in config)

DEMO – USING MOBIUS C# SHELL

sparkclr-shell.cmd

```
> var rdd = sc.Parallelize(Enumerable.Range(0, 100), 2);
> rdd.Reduce((x,y) => x+y) //prints sum of the values;
```



DEMO – USING F# SHELL

fsi.exe

```
> sparkclr-submit.cmd debug
```

```
> fsi --use:C:\temp\mobius-init.fsx

let dataframe =
session.Read().Json(@"C:\temp\data.json");;

dataframe.Show();;
dataframe.ShowSchema();;
dataframe.Count();;
```

Kafka Streaming Example

```
StreamingContext sparkStreamingContext = StreamingContext.GetOrCreate(checkpointPath,
    () =>
    {
        var ssc = new StreamingContext(sparkContext, slideDurationInMillis);
        ssc.Checkpoint(checkpointPath);

        var stream = KafkaUtils.CreateDirectStream(ssc, topicList, kafkaParams, perTopicPartitionKafkaOffsets);
        var countByLogLevelAndTime = stream
                    .Map(kvp => Encoding.UTF8.GetString(kvp.Value))
                    .Map(line => line.Split(',')) //message format:[timestamp],[loglevel],[logmessage]
                    .Map(columns => new KeyValuePair<string, int>($"{columns[0]},{columns[1]}", 1))
                    .ReduceByKeyAndWindow((x, y) => x + y, (x, y) => x - y, windowDurationInSecs, slideDurationInSecs, numPartitions)
                    .Map(logLevelCountPair => $"{logLevelCountPair.Key},{logLevelCountPair.Value}");

        countByLogLevelAndTime.ForeachRDD(countByLogLevel =>
        {
            countByLogLevel.SaveAsTextFile($"{appOutputPath}/{Guid.NewGuid()}");
        });

        return ssc;
    });

sparkStreamingContext.Start();
sparkStreamingContext.AwaitTermination();
```

Initialize StreamingContext & Checkpoint

Create Kafka DStream

Use DStream transformations to count logs by loglevel within a time window

Save log count

Start stream processing

Mobius in Linux

- Mono is used for using Mobius with Spark in Linux
- Mobius project CI (build, unit & functional tests) in Ubuntu
- Mobius validated in Ubuntu, CentOS, OSX
- Mobius validated with Spark clusters in Azure HDInsight and Amazon Web Services EMR
- More info at linux-instructions.md @ GitHub

Project Info

- https://github.com/Microsoft/Mobius
  Contributions welcome!

- MIT license

- Discussions
  - StackOverflow: tag "SparkCLR"
  - Gitter: https://gitter.im/Microsoft/Mobius
  - Twitter: @MobiusForSpark

CSharpRDD

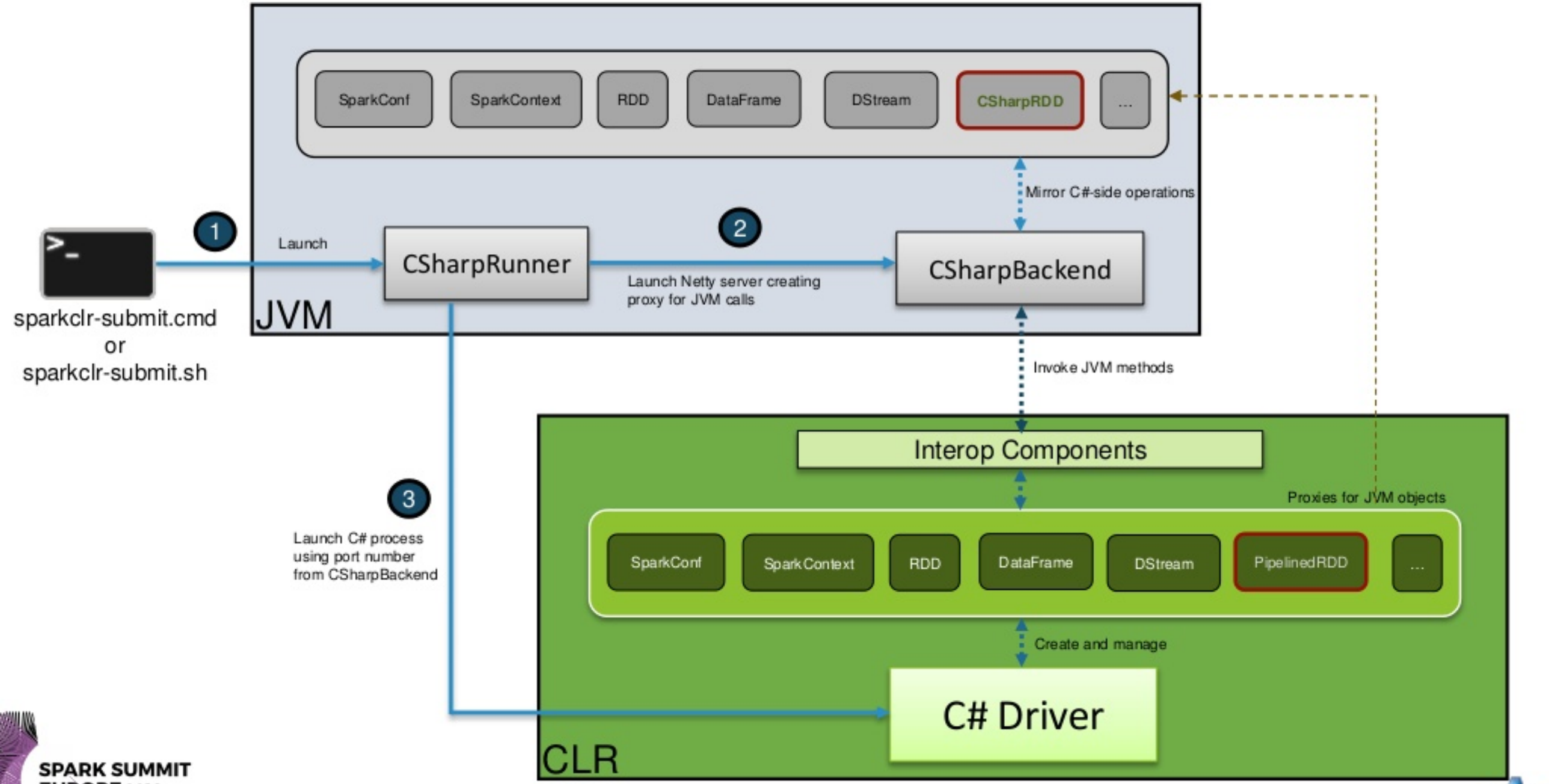
- C# operations use CSharpRDD which needs CLR to execute
  - If no C# transformation or UDF, CLR is not needed ~ <u>execution is entirely JVM-based</u>
- RDD<byte[]>
  - Data is stored as serialized objects and sent to C# worker process
- Transformations are pipelined when possible
  - Avoids unnecessary serialization & deserialization within a stage

Performance Considerations

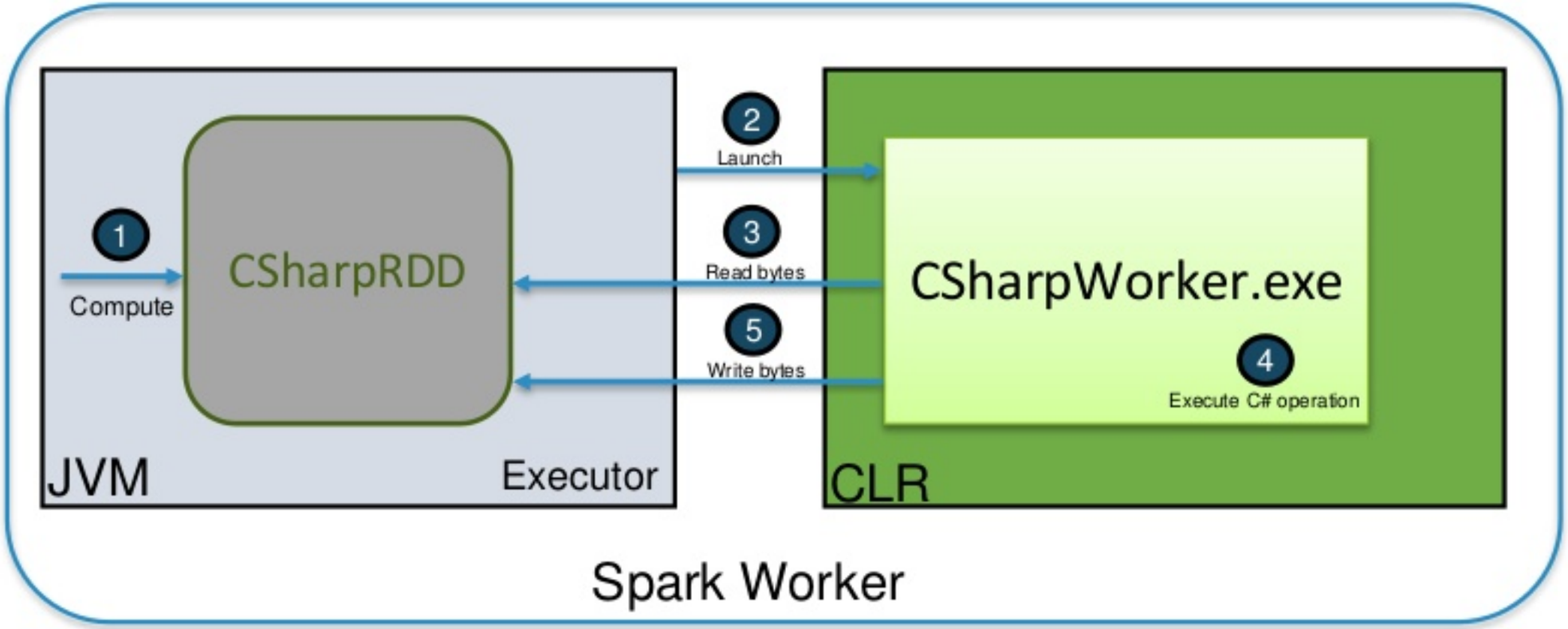
- Map & Filter RDD operations in C# require serialization & deserialization of data ~ impacts performance
  - C# operations are pipelined when possible ~ minimizes Ser/De
  - Persistence is handled by JVM ~ checkpoint/cache on a RDD impacts pipelining for CLR operations

- DataFrame operations without C# UDFs do not require Ser/De
  - Perf will be same as native Scala-based Spark application
  - Execution plan optimization & code generation perf improvements in Spark leveraged

INTERNALS OF DRIVER & WORKER


SparkConf
SparkContext
RDD
DataFrame
DStream
CSharpRDD
...
Mirror C#-side operations
1
Launch
CSharpRunner
2
Launch Netty server creating proxy for JVM calls
CSharpBackend
JVM
sparkclr-submit.cmd
or
sparkclr-submit.sh
Invoke JVM methods
Interop Components
Proxies for JVM objects
3
Launch C# process using port number from CSharpBackend
SparkConf
SparkContext
RDD
DataFrame
DStream
PipelinedRDD
...
Create and manage
C# Driver
CLR

Worker-side Interop



THANK YOU.

- Mobius is production-ready & Cloud-ready
- Use Mobius to build Apache Spark jobs in .NET
- Contribute to github.com/**Microsoft/Mobius**
- @MobiusForSpark

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